Force, Motion and Energy

Distance and Displacement

- Distance and displacement both describe the amount a body has moved from one location to another.
- Distance is the length of the path the body follows. It is measured in metres (m)
- Displacement takes into account the *direction* travelled. It is the distance between the initial and final positions as measured in a straight line. It is also measured in metres.
- In an athletic event such as the 400m, the distance travelled is not the same as the displacement due to it starting and finishing at the same place!

Activity

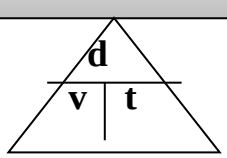
Handout: Distance & Displacement

OHP: Class exercise: Mr Martins Trip

Speed & Velocity

- The rate at which a body moves from one location to another is usually described as either speed (s) or velocity (v)

$$\textbf{Average velocity (ms}^{-1}\textbf{)} = \frac{\textbf{Distance covered (m)}}{\textbf{Time taken (s)}}$$



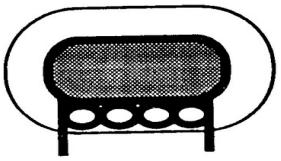
- With reference to the last exercise; if it took 96 hours to travel from Seattle to New York (non-stop) calculate average speed as kmh^{-1} ($48kmh^{-1}$)
- Egs:
  - What is the speed of a car in kmh^{-1} if it covers 15km in 30 minutes?
  - Find the speed of a runner in ms^{-1} if she covers 1000m in 7 minutes.

Converting Units

kmh^{-1} is converted to ms^{-1} by…… x 1000 (m in km)
÷ 3600 (s in hr)

ms^{-1} is converted to kmh^{-1} by ….. x 3600
÷1000

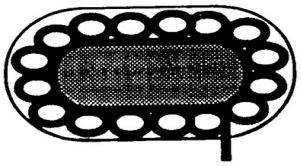
Displacement equals Distance



1. 100m 2. 3.

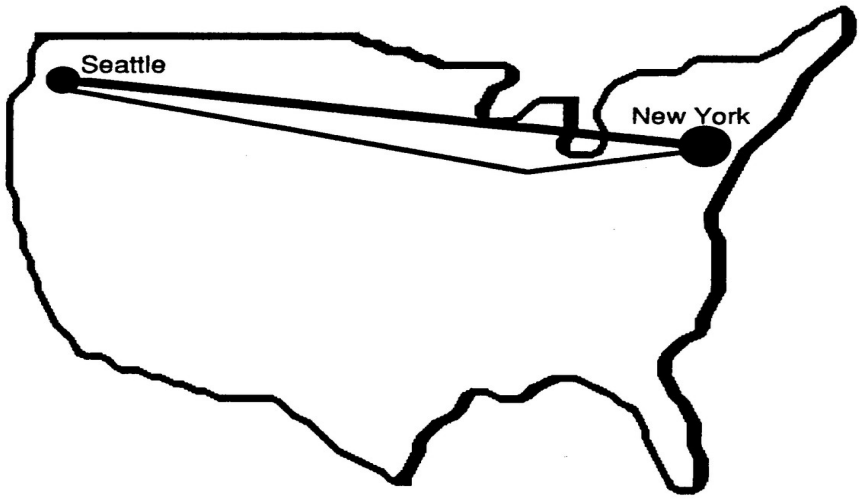
Start Finish

Displacement differs from distance



1. 400m 2. 3.

Start / Finish



Distance ——————

Displacement ▬▬▬▬▬

Travelling from Seattle to New York

Distance = 4600 km
Displacement = 4000 km

Remembering that displacement is the distance between the initial and the final position, a round trip from Seattle / New York / Seattle would then be:

Distance = km
Displacement = **km**

- Egs:
- A cricket ball is bowled at 134km per hour.
  - Convert 134 kmh^{-1} to ms^{-1} *(37.22)*
  - If the batsman is 18 m from the bowler when the ball is released, how long has he to prepare this stroke? *(about 0.5 s)*
- Stephen fancied himself as a tennis player. He asked Peter to time his serves from the serve line to the netting fence (30m)

  - Find the speed of the serve in metres per second if the measured time was 1.2s. (*25)*
  - Convert this speed to kmh^{-1}

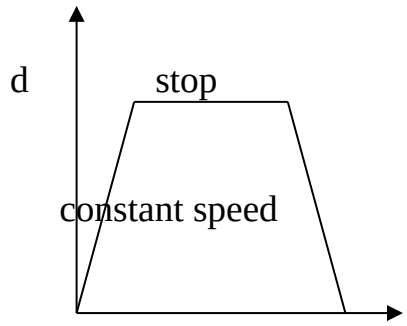
Activity:
Motoring Madness p67 Yellow Practical manual

Measuring Speed in the Laboratory p70-72 (Get results as a demo) – Before doing speed-time graph, teach acceleration (below)

*Give them some fake ticker timer results and ask the trend ie constant, acc, dec

Hwk: A Toy car Journey p73-4 Yellow Practical manual

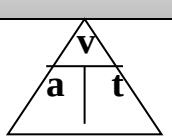
Distance-Time Graphs



Acceleration

- An object changing its speed is accelerating – positively or negatively (decelerating)

$$\text{Acceleration } (ms^{-2}) = \frac{\text{final velocity} - \text{initial velocity } (\Delta v)}{\text{Time taken (s)}}$$

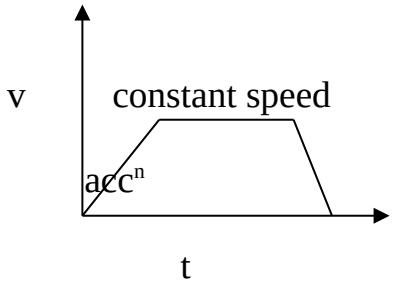


- Eg: Calculate acceleration of a trolley which changes from $2ms^{-1}$ to $5ms^{-1}$ in 2 seconds.

$a = 5-2 \div 2 = 1.5ms^{-2}$

Activity:
Finish speed-time work in practical book p72

Velocity-Time graphs



Activity:
Pathfinder: p38-9 no's 3, 4, 5, 6, 7, 10,11

Acceleration Activity

Homework: Handout: Quick Reflexes

Handout: Worksheet on "Falling"

Quick reflexes

Dien and Arapeta conducted an experiment to measure the reaction times of the thirty students in their class.

Dien stood beside the student, who was seated, holding a one metre ruler above his or her outstretched hand. He dropped the ruler, without warning, and the student caught it between his or her fingers (see the photograph). Arapeta recorded the distance the ruler fell before it was caught, on the scatter diagram shown below.



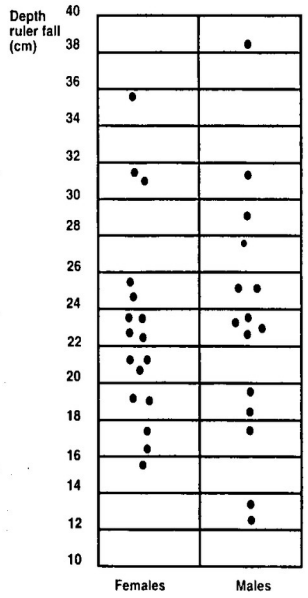
Terry Wales

1 On the scatter diagram, circle the individual with the fastest reaction time.

2 Complete this table summarising the results of the experiment.

Depth of ruler fall (cm)	Frequency (males)	Frequency (females)
12–16		
16–20		
20–24		
24–28		
28–32		
32–36		
36–40		

Scatter diagram of depth of fall of ruler for males and females



3 Convert a speed of 60 km/h into m s^{-1}.

4 A fall of 24 cm corresponds to a reaction time of 0.22 s. Calculate the distance, in metres, travelled by a car moving at a speed of 60 km/h in this time.

5 According to road safety manuals, an average driver travelling at 60 km/h will travel 12 m before they are able to put on the brakes for an emergency stop. Calculate this reaction time.

6 Account for the differences in reactions times for questions 4 and 5.

7 A car travelling at 60 km/h takes 2.88 s to stop.

a Calculate the change in speed, in m s^{-1}.

b Calculate the deceleration of the car in m s^{-2}.

c Calculate the stopping force on the car if it has a mass of 1025 kg.

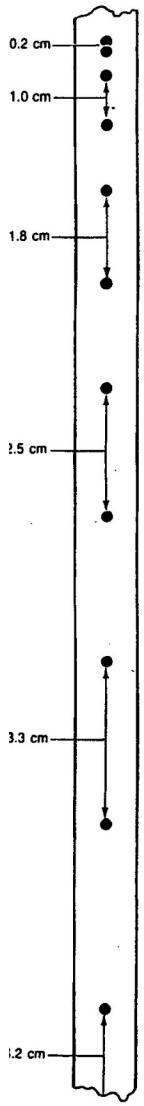
Acceleration

Method:

1. Form groups of 6
2. Choose one member of your group to run 50 metres at maximum speed
3. The remainder of the group collect a stopwatch each and place themselves at every 10 metres.
4. On the “go” signal, the runner starts sprinting around the course at maximum speed and all the timers start their stopwatches. As the runner passes each recording position, the timer at that mark stops his/her watch and records the time the runner passed their station
5. Fill in the results section and work through the questions

Displacement	Time(s)	Velocity for 10m interval (ms^{-1})	Final-Initial velocity	Acceleration (ms^{-2})
10m				
20m				
30m				
40m				
50m				

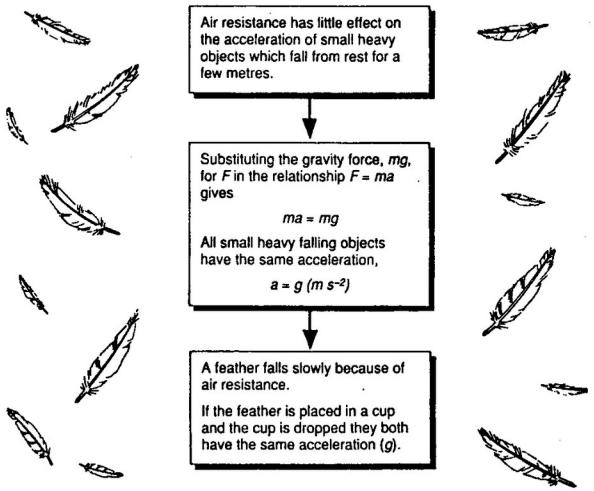
Construct a velocity-time graph and using the graph, describe the acceleration. Can you calculate the acceleration from the graph?



FALLING

Constant acceleration

FALLING IN AIR



Air resistance has little effect on the acceleration of small heavy objects which fall from rest for a few metres.

Substituting the gravity force, mg, for F in the relationship $F = ma$ gives

$ma = mg$

All small heavy falling objects have the same acceleration,

$a = g\ (m\ s^{-2})$

A feather falls slowly because of air resistance.

If the feather is placed in a cup and the cup is dropped they both have the same acceleration (g).

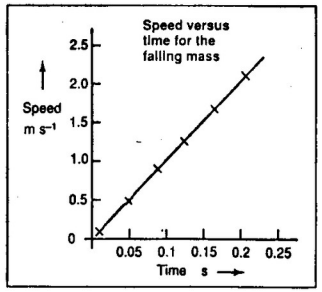
ACCELERATION DUE TO GRAVITY

Figure 7.1 is a copy (actual size) of a ticker-timer tape which Craig attached to a 1 kg mass which he then dropped. His speed-time data is plotted as a graph in 7.2.

Results

Separation of dots (cm)	Time (s)	Speed ($m\ s^{-1}$)
0.2	0.01	0.1
1.0	0.05	0.5
1.8	0.09	0.9
2.5	0.13	1.25
3.3	0.17	1.65
4.2	0.21	2.10

7.2



The speed-time graph is a straight line; the acceleration is constant. The slope of the graph gives the acceleration, due to gravity, g, in metres per second per second.

From the graph... [g = 10 m s^{-2}]

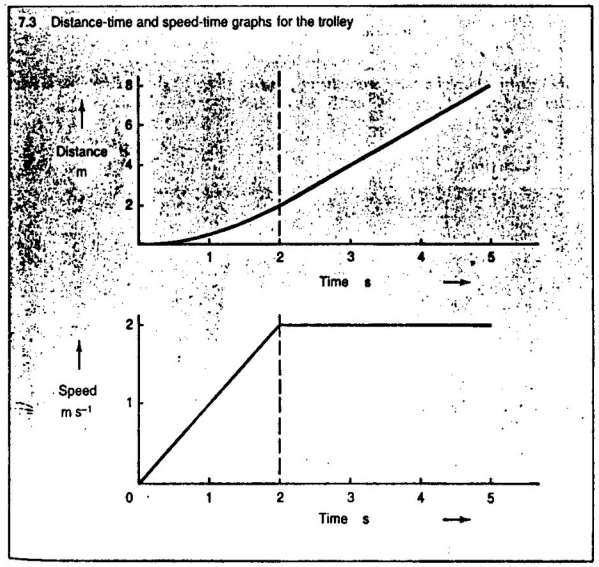
NOTE: more accurate measurements give g as 9.80 m s^{-2}. g is often taken as 10 m s^{-2} which leads to only a 2% error in simple calculations.

CONSTANT ACCELERATION

Distance-time graph

A trolley with well-oiled wheels accelerates down a straight ramp and rolls across a level floor at constant speed. The distance-time graph is made up of two parts. A curved line when the trolley is accelerating down the ramp and a straight line as it crosses the floor.

7.3 Distance-time and speed-time graphs for the trolley



Speed-time graph

The acceleration on the ramp is constant because gravity provides a constant force. The speed-time graph for the journey is made up of two parts. A straight line when the trolley is on the ramp with constant acceleration and a second straight line when it rolls across the floor at constant speed.

GRAPH STUDY

1 Look carefully at the distance-time graph. Write down:

a The length of the ramp.
b The time that the trolley was on the ramp.
c The total time for the journey.
d The total distance covered.

2 The same trolley runs down a second ramp which is longer and steeper. It reaches the floor at the same time.

Sketch a possible distance-time graph for the second journey on the same axes.

3 Look carefully at the speed-time graph. Write down:

a The maximum speed.
b The time for which the acceleration is zero.
c The slope of the graph during the first two seconds.
d The acceleration during the first two seconds in m s^{-2}

4 The same trolley runs down a third ramp which is shorter and less steep. It reaches the floor at the same time.

Sketch a possible speed-time graph for the third journey on the same axes.

Mass and Weight

- Astronauts are weightless in space but they're not thinner! The amount of matter in their body hasn't changed. This is their mass (kg)
- Weight is a measure of the force of gravity on a mass and is measured in Newtons (N)

Weight = mass *g
$F_w = m.g$

- Gravity forces objects to accelerate towards centre of earth at $10ms^{-2}$ or $10Nkg^{-1}$
- Weight changes as you move away from the earth's surface
- The force of gravity on the moon is 1/6 of that on the moon

Questions:
1. Jill has a mass of 54 kg. What is her weight in Newtons on Earth? What about on the moon?
2. What is the approximate force due to gravity of the following masses:
   6kg 600g 4g 72kg
   on earth? On the moon?

Activity:
Yellow Practical Book: p79 A Weighty Problem

Forces

- A push, pull or twist which can change an objects motion
- The unit of force is the Newton (N)

Activity:
Blue book p49 : Answer the following questions as notes.
1. What are action and reaction forces?
2. What are balanced and unbalanced forces?
3. How do you calculate the unbalanced force?

Momentum

Activity:
Planning Exercise: Collisions p75 Yellow Prac book –HOMEWORK
Carry out plan then answer questions p78
Demo:

Spin a fresh egg and stop it spinning and repeat with a boiled egg, is there a difference?

Force, Mass and Acceleration

- Newton's Second Law of Motion
- When a body is acted on by force…
  - The greater the force the greater the acceleration
  - The smaller the mass, the greater the acceleration
  - The change in motion takes place in the direction in which the force is applied

Force(N) = mass (kg)* acceleration(ms^{-2})

F= m.a

NB: Unit can also be $kgms^{-2}$

Activity:

Aliens OHP worksheet
Step on it! Worksheet
May the Force be with you p81-2 Yellow Practical book

Compare slopes of velocity time graphs if force is changed ie increased force is increased gradient

Compare slope of velocity time graphs if mass is changed ie increased mass is decreased gradient.

Activity:

Pathfinder p42-3 no's 4,6,7,8,10,

STEP ON IT!

Acceleration, force and mass

FORCE AND ACCELERATION

Almost any car will pull away from traffic lights much faster than a truck and as every teenager knows a small car with a powerful engine will leave a standard car behind. For maximum acceleration a large powerful engine must be coupled with a small overall mass. For this reason a large motor bike will out-accelerate anything else on the road.

PUSHING TROLLEYS

Several people push a heavy free-running baggage trolley which accelerates on level ground. The results are tabulated below.

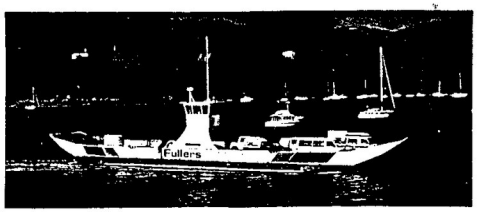
The acceleration depends on both the mass of the trolley and the unbalanced force applied to it.

1 Copy the table and fill in what you would expect the missing entries to be.

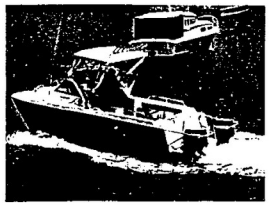
Force (Number of people pushing) (N × 100)	Mass (Bags plus trolley) (kg × 100)		Acceleration (Metres per second per second) ($m\ s^{-2}$)
1	1		1
2	1		2
1	2		?
2	4		?
?	2	?	4

2 Describe the pattern you have used to predict the values you have added.



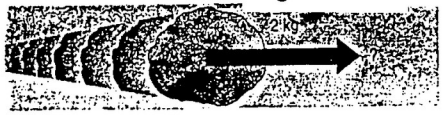


6.1 a A ferry picks up speed very slowly because of its very large mass.



b A tiny boat with a big engine takes off like a rocket!

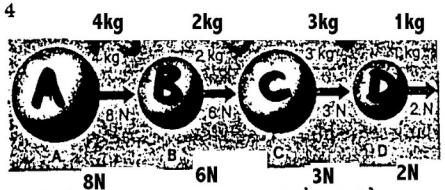
2kg mass



What force is needed to make the rock accelerate at
a $2\ m/s^2$ b $0.5\ m/s^2$ c $4\ m/s^2$?

3 In an experiment with a 0.5 kg trolley, someone measures the pulling force, calculates the acceleration, writes down the values '10' and '5', but forgets to note which is which.
Can you decide, and add the correct units?

4



a Which masses have the same acceleration?
b Which mass has most acceleration?
c Which mass has least acceleration?

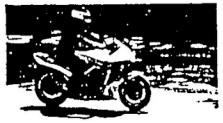
5 The three vehicles below are among the most powerful of their type.
Which wins the contest for the best acceleration?
Which is the loser?



Boeing 747:
mass 400 000 kg
force from engines 800 000 N



Porsche 911:
mass 1300 kg
force from engine 7800 N



Honda 1000
mass 300 kg
force from engine 3000N

ACCELERATION AND UNBALANCED FORCES

$F = ma$

1 A car with a mass of 1200 kg is accelerating on a level straight road at 4 $m\ s^{-2}$. Find the unbalanced force applied to the car by the engine.

2 A bullet (mass 20 g) is accelerating at 200 000 $m\ s^{-2}$ in the barrell of a rifle. Find the unbalanced force exerted on the bullet by the expanding gasses.

3 A catapult applies an unbalanced force of 40 newtons to a stone of mass 0.25 kg.

a Show that $a = \frac{F}{m}$

b Find the acceleration of the stone.

4 An unbalanced force of 200 newtons accelerates a baggage trolley along a level platform at 2.0 $m\ s^{-2}$.

a Rearrange $F = ma$ to obtain m in terms of F and a.
b Find the total mass of the loaded trolley

5 A car ferry engine provides a total thrust of 5000 newtons to a 100 000 kg ferry.

a Find the acceleration of the ferry if the water were to provide no resistance.
b The measured acceleration of the ferry is 0.01 $m\ s^{-2}$. Find the actual unbalanced force acting on the ferry.
c Write down the drag on the hull due to the water.

$F = ma$

Plan and complete your own investigation.

Attempt to show . . .

EITHER: that the acceleration of a given mass is proportional to the applied force.

OR: that the acceleration for a given force is halved if the mass is doubled.

KEY RELATIONSHIP

$F = ma$

This important relationship is known as Newton's second law.

Energy

- Law of Conservation of Energy
  Energy is neither created nor destroyed, it just changes form"
  - Lightbulb changes from electrical to….
  - Hydrodam changes from kinetic to….
- When something happens, energy is transferred
- Energy is needed to carry out work. When we do work, we use up energy
- All calculations involving energy use the unit joule (J) or kilojoule (kJ)

Work

- This is done when a FORCE moves an object over a DISTANCE

Work (J) or (Nm) = Force (N) * Distance (m)
W= F.d

- The amount of work done is a measure of the energy transferred
- Eg A force of 8N is exerted on a body while it moves 3m. How much work is done?

  W=F.d
  =8*3
  =24N
- Something which does work LOSES energy.
- Something which has work done on it GAINS energy
- So… if I push a box across a floor, the amount of energy I lose should equal the amount of energy the box gains. This doesn't usually happen due to the friction forces between the box and floor
- Frictional forces oppose movement. Often, heat is produced eg braking in cars, bikes.

Activity:
Work OHP

Work examples

1. **12J of energy is transferred to a mass while it moves 0.4m. What force was exerted on the mass?**

2. **If a mass of 4kg is accelerated at $3ms^{-2}$ for a distance of 2m, how much work is done?**

3. **A boy loses 60J of energy when he pushes a sled 5m across grass. The energy gained by the sled is 50J**

   a) **what has happened to the rest of the energy lost by the boy?**
   b) **What must the force of friction be?**

POWER

A skier has a mass of 60kg. It takes her 40 s to climb 20m vertically to the top of a slope

Calculate: a) **her weight**
b) **work done when she climbs the slope**
c) **average power output**

Why in reality would she have to do more work than has been calculated?

Power

- Power is the rate at which work is done or energy is changed

Power (W) or (Js^{-1})= Work done (J) = F*d
Time (s) t

- Eg what is the power output of a woman who exerts a force o 50N to shift a chair 1m in 5 s? P = Work ÷time
  = 50*1÷5
  = 10W
- Eg A woman cycles along the road and does 1800J of work in one minute. What is her power output? P= Work÷time
  = 1800÷60
  = 3W

Finish OHP question

Activity:
Body Power p85-7 Yellow Practical Book

Gravitational Potential Energy (E_p)

- When we do work on something we transfer energy to it
- So.. if we lift an object, we have done work and thus LOST energy which is transferred and STORED in the object after it has been lifted off the ground
- If an object falls, its stored E_p is changed to other forms of energy such as sound, kinetic, heat
- An object on the ground has zero gravitational potential energy so the height an object is lifted is important

E_p (J) = m.g.Δh
= mass* force due to gravity*change in height (final-initial)

Note: (m*g) (Δh) is the same as F*d
Force due to gravity is $10Nkg^{-1}$

- Calculate E_p when a 50kg sack of flour is lifted to a table 0.6m in height
  E_p = 50*10 *(0.6-0)
  = 500 * 0.6
  = 300Nm ⇒ 300J (The sack of flour has gained 300J)
- Note that if you word this example as "How much work is done when lifting a 50kg sack of flour 0.6m?" , the same answer is achieved

Fit for the game

Leon and Barry are on the school rugby team. They both jog to keep fit.

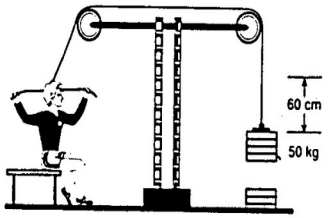
1 Leon jogs for 30 minutes at an average speed of 3 m s^{-1}. Barry jogs 5 km in 25 minutes.

a How far does Leon jog?

b What is Barry's average speed?

The boys also use the weights room at the gym.

2 Leon uses the lat pull. By pulling down on the handle bar, he lifts up the weights.



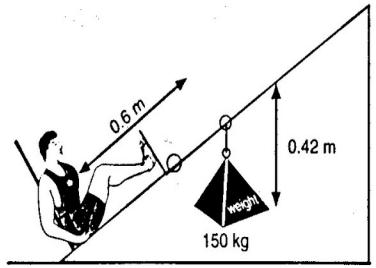
a What force must he pull with to lift the 50 kg weight?

b He raises the weights 60 cm. Calculate the energy gained by the weights.

c The weight is lowered to its starting position at a constant speed. What is the force applied?

d Leon raised the 50 kg weight in 1.2 s. Calculate his power.

3 Barry uses the leg press. He pushes a 150 kg mass 0.6 m up the incline, as shown in the diagram. Calculate the work he has done.



4 On the rugby field Leon and Barry accidentally collide with each other while running to catch a high ball. Leon has a mass of 70 kg and was running at a speed of 3.5 m s^{-1}. Barry has a mass of 80 kg and was running at a speed of 3.0 m s^{-1} in exactly the opposite direction to Leon.

a Calculate the momentum of each boy.

b Which boy will be knocked backwards?



Barry
80 kg
3.0 ms-1



Leon
70 kg
3.5 ms-1

Gravitational Potential Energy

1. **What is the E_p of a 10kg block of chocolate (yumyum!) stored on top of a 3m high cupboard?**

2. **What is the E_p of 2kg apples stored in a shopping bag on a shelf 750mm off the ground?**

3. **What is the E_p of a 600g bird gliding at 40m?**

4. **What is the change in E_p when a 1.5kg packet of sugar falls from a shelf 1.9m above the ground and lands on the bench 1.3m above the ground?**

5. **A builder on a 10m high roof drops a 1kg tile which lands on a deck 150cm above the ground? What is the change in E_p?**

Kinetic Energy

- This is the energy of moving objects
- Kinetic Energy (E_k) is related to mass and speed
- If 2 elephants have the same mass but different speeds, the faster elephant has more E_k
- If 2 elephants have the same speed but different masses, the heavier elephant has more E_k

$E_k = ½mv^2$
$= ½*mass*(velocity^2)$
Unit is JOULES

- Eg What is the kinetic energy of a 1000kg car travelling at $20ms^{-1}$?

$E_k = ½ * 1000* 20^2$
$= 500 * 400$
$= 200000Nm \Rightarrow$

Activity
Pathfinder p46-7 no's 3,4,7,8,10